

Econ 440: Statistics Review

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In order to understand the concepts and methods that we will discuss in this class, it is important to have an underlying understanding of some key concepts from statistics.

1 Parameters and Estimators

Definition. A *parameter* is an object (e.g., a quantity) from a given population that we want to learn about.

- Examples:
 - Average earnings of all US people over 18 years old
 - Median education of all California adults
 - Average Body Mass Index (BMI) of all US people
 - Average effect of attending college on earnings for an adult in the US
- Notation: Let the parameter (e.g., average earnings of all US people over 18 years old) be denoted by α .
- Issue: α is usually unknown.
 - For example, for the first three cases above, it may be impossible, impractical, or very costly to compute α .
- Goal: Estimate (or “guess”) the value of α .
- Usual approach: We get a *random sample* size n of the population of interest, and use it to make *inferences* about the population (say, about α).
 - Remember that a random sample is one where the observations are *independent and identically distributed (i.i.d.)*.

- Example: Let α = Average earnings of all US people above 18 years old, and let Y_i = the earnings of individual i .

- Idea: If n is large enough $\implies$ The sample will be representative of the population (e.g., approximately same fraction of females, males, Whites, Blacks, same average education, age, etc.) $\implies$ Use the *sample average* $\bar{Y}$ (i.e., the average *in* our sample) to estimate the population or “true” average α .
- Intuition: If sample is “similar” to the population $\implies$ we expect $\bar{Y}$ to be “similar” to α . In fact, the *Law of Large Numbers (LLN)* states that as $n \rightarrow \infty$ (say, as our sample size gets larger), the probability that $\bar{Y}$ is arbitrarily close to α approaches one.

Definition. A function of the data in our sample (called an “*statistic*”) that is used to make inferences about a population parameter α is called an *estimator* of α , and is usually denoted by $\hat{\alpha}$. An *estimate* is the specific value of $\hat{\alpha}$ in a particular sample.

- Example: The *estimator* of α in the example above is $\hat{\alpha} = \bar{Y} = n^{-1} \cdot \sum_{i=1}^n Y_i$. Suppose that we had a sample size $n = 5$ with values $\{80, 100, 400, 280, 140\}$. Then, our *estimate* of α would be $\hat{\alpha} = \bar{Y} = 200$.

- **Key idea:** The parameter α is a *fixed* quantity, i.e., it is not random. However, the estimator $\hat{\alpha}$ is random; in other words, it has a distribution.

Why is $\hat{\alpha}$ random? Because its value depends on the specific sample we draw. If we were to take another sample $\implies$ we would get a different estimate/value.

Example 1:

Example 2: Suppose we flip a fair coin, and let $X = 1$ if the coin is heads and $X = 0$ if it is tails. Let $\alpha = Pr(X = 1)$, i.e., the probability that the coin is heads. (Note, this is a case where the parameter is known: $\alpha = 1/2$.)

2 Expectations or Means

In this section, we formally define what a population or “true” (or “mathematical”) average is and discuss some of its properties that we will use throughout the course. The “true” or mathematical average of a random variable X is called the *expectation* or *mean* of X , and we denote it by $E(X)$.

2.1 Expectations or Means: Definition

Let X be a *random variable* that can take values in $\mathcal{X} \in \mathbb{R}$, where $\mathcal{X}$ is the space of X , that is, the set of all possible values that X can take.

- Example A: In Example 2 above, where we flip a fair coin and let $X = 1$ if the coin is heads and $X = 0$ if it is tails, $\mathcal{X} = \{0, 1\}$.
- Example B: If X is a normally distributed random variable, then $\mathcal{X} = \mathbb{R}$.

Following the usual notation, in this class we use capital letters (e.g., X) to denote a random variable, and lower case letters (e.g., x) to denote particular values (e.g., 5, 8, etc.).

Definition. A random variable X is *discrete* if it can only take a finite (or at most a countable) number of values. A random variable X is *continuous* if it can take a continuum or uncountable number of values.

- Examples: In Example A above, X is a discrete random variable, while in Example B X is continuous.

We now formally define the expectation or mean of a random variable.

Definition. If X is a discrete random variable that can take values in $\mathcal{X}$, its *Expectation*, *Mean*, or *Expected Value* is given by:

$$E(X) = \sum_{x \in \mathcal{X}} x \cdot f(x), \quad (1)$$

where $f(x) = \Pr(X = x)$ is called the *probability mass function* (pmf) of X , and equals the probability that X equals a given value x .

If X is a continuous random variable that can take values in $\mathcal{X}$, its *Expectation*, *Mean*, or *Expected Value* is given by:

$$E(X) = \int_{x \in \mathcal{X}} x \cdot f(x) dx, \quad (2)$$

where $f(x)$ is the *probability density function* (pdf) of X .

Intuition: If X is discrete, the expectation of X , $E(X)$, is calculated by summing over all possible values of X times the probability of observing each particular value of X :

$$E(X) = \sum_{x \in \mathcal{X}} x \cdot f(x) = \sum_{x \in \mathcal{X}} x \cdot \Pr(X = x)$$

$\implies E(X)$ is a *weighted average*, where the weights are given by $f(x) = \Pr(X = x)$.

A similar intuition applies to the case when X is continuous, with sums replaced by integrals and the pmf replaced by the pdf (remember: integrals are like sums for the continuous case).

Examples:

- *Example 1:* Suppose we flip a coin, and let:

$$X = \begin{cases} 1 & \text{if heads} \\ 0 & \text{if tails} \end{cases}$$

Then, X is a discrete random variable, with $\mathcal{X} = \{0, 1\}$. The pmf of X is given by:

$$f(x) = \begin{cases} \Pr(X = 0) = \frac{1}{2} \\ \Pr(X = 1) = \frac{1}{2} \end{cases}$$

Now, we obtain the expected value of $E(X)$ using the definition of $E(X)$ in (1):

$$E(X) = \sum_{x=0}^1 x \cdot f(x) = 0 \cdot \Pr(X = 0) + 1 \cdot \Pr(X = 1) = 0 \cdot \frac{1}{2} + 1 \cdot \frac{1}{2} = \frac{1}{2}$$

$$\implies \boxed{E(X) = \frac{1}{2}}$$

Note: This is the Expected Value (or the “true” mean) of X . If we flipped a coin 10 times, we may not get 5 heads and 5 tails. Hence, the *sample average or mean* may not be equal to $\frac{1}{2}$. However, if we were to flip a coin an infinite number (∞) of times, we would *expect* to get heads $\frac{1}{2}$ the time.

- *Example 2:* Suppose X followed a Normal Distribution with mean μ and variance σ^2 , i.e., $X \sim N(\mu, \sigma^2)$. In this case, X is a continuous random variable with $\mathcal{X} = \mathbb{R}$. The pdf $f(x)$ of a normally distributed random variable $X \sim N(\mu, \sigma^2)$ is given by:

$$f(x) = (2\pi\sigma^2)^{-1/2} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right)$$

It can be shown that (i.e., you do not have to do the algebra), by using the definition of the expected value of $E(X)$ in (2), we obtain:

$$E(X) = \int_{x \in \mathcal{X}} x \cdot f(x) dx = \int_{-\infty}^{\infty} x \cdot (2\pi\sigma^2)^{-1/2} e^{\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)} dx = \mu$$

Hence, as we know, the expected value or mean of a normally distributed random variable X equals μ .

Interpretation: Some ways to interpret $E(X)$:

- $E(X)$ is a weighted average of X , with weights given by $f(x)$.
- $E(X)$ is a “long run” average, that is, the average of X that we would obtain if we could get an infinite random sample—one with “ $n = \infty$ ”—from the population (i.e., if we could sample everybody in our population of interest), or if we could repeat the experiment (e.g., flipping a coin) an infinite number of times.

- *Example 1:* In Example 1 above (the “flipping a coin example”), if we were to flip the coin an infinite number of time ($n = \infty$), we would expect the sample average to equal $1/2$. Informally,

$$\lim_{n \rightarrow \infty} \frac{\sum_{i=1}^n X_i}{n} = \frac{1}{2} = E(X)$$

- *Example 2:* Consider Example 1 from the end of Section 1, where we let Y = earnings, and our population of interest was all US people older than 18 years old. Then:

Note: Remember, the expectation or “true” mean $E(Y)$ is *different from* the sample average $\bar{Y} = n^{-1} \cdot \sum_{i=1}^n Y_i$. The first one is not random (i.e., is fixed), while the second is random as its value depends on the specific sample we got (see discussion at the end of Section 1).

2.2 Conditional Expectations or Means

The conditional expectation or mean of a random variable Y conditional on another random variable X gives the expectation or “true” mean of Y given that X took the particular value x . We write it as: $E(Y|X = x)$.

- For example, the expected value of earnings given that the individual attended college; or the expected earnings for females.

Definition. If Y is a discrete random variable, the *conditional expectation or mean* of Y given $X = x$ is given by:

$$E(Y|X = x) = \sum_{y \in \mathcal{Y}} y \cdot f(y|X = x), \quad (3)$$

where $f(y|X = x)$ is the conditional pmf of Y given X , that is, the probability that $Y = y$ given that $X = x$: $f(y|X = x) = \Pr(Y = y|X = x)$.

If Y is a continuous random variable, the same definition above applies with the summation being replaced by integration, and the conditional pmf being replaced by the conditional pdf.

Example: Suppose we flip two fair coins. Define the following random variables:

$$\text{Let } X = \begin{cases} 1 & \text{if 1st flip is heads} \\ 0 & \text{if 1st flip is tails} \end{cases}$$

Let Y = Total Number of Heads out of 2 Flips.

In this particular problem, we focus on the random variable Y .

- *First, as an exercise and for comparison purposes, we obtain the (unconditional) expectation of Y : $E(Y)$.*

To use the formula in (1) to obtain $E(Y)$, we first need to obtain the pmf (or distribution) of Y , $f(y)$. To do so, we first list the possible outcomes of this experiment (let H denote “heads” and T “tails”):

Possible Outcomes of 2 Flips: $\{(H, H), (H, T), (T, H), (T, T)\}$.

Hence, the pmf of Y is given by: $f(y) = \begin{cases} \Pr(Y = 0) = \frac{1}{4} \\ \Pr(Y = 1) = \frac{1}{2} \\ \Pr(Y = 2) = \frac{1}{4} \end{cases}$

Then, using the definition of the expectation $E(\cdot)$ in (1) we find:

$$\begin{aligned} E(Y) &= \sum_{y=0}^2 y \cdot f(y) = 0 \cdot \Pr(Y = 0) + 1 \cdot \Pr(Y = 1) + 2 \cdot \Pr(Y = 2) = \\ &= 0 \cdot \frac{1}{4} + 1 \cdot \frac{1}{2} + 2 \cdot \frac{1}{4} = 1 \\ &\implies \boxed{E(Y) = 1} \end{aligned}$$

Therefore, the expected number of heads out of two flips of a coin is $E(Y) = 1$.

- Now, we calculate the conditional expectation of Y given that the first flip is heads (i.e., given $X = 1$), or $E(Y|X = 1)$.

To use the formula to obtain $E(Y|X = 1)$ in (3), we first need obtain the *conditional pmf* (or conditional distribution) of Y given X , $f(y|X = x)$. To do so, we first list the possible outcomes of this experiment *given that the first flip was heads, i.e., given $X = 1$* :

Possible Outcomes if First Flip is Heads $\{(H, H), (H, T)\}$.

Hence, the conditional pmf of Y given that $X = 1$ is given by:

$$f(Y|X = 1) = \begin{cases} \Pr(Y = 1|X = 1) = \frac{1}{2} \\ \Pr(Y = 2|X = 1) = \frac{1}{2} \end{cases}$$

Then, using the definition of the conditional expectation in (3) we find:

$$\begin{aligned} E[Y|X = 1] &= \sum_{y=1}^2 y \cdot f(y|X = 1) = 1 \Pr(Y = 1|X = 1) + 2 \Pr(Y = 2|X = 1) = \\ &= \frac{1}{2} + 2 \cdot \frac{1}{2} = \frac{3}{2} \\ &\implies \boxed{E[Y|X = 1] = \frac{3}{2}} \end{aligned}$$

Therefore, the expected number of heads out of two flips of a coin *given that the first flip was heads* (i.e., given $X = 1$) is $E(Y|X = 1) = 3/2$.

Note that, as one would expect, the expected number of heads given that the first flip was heads, $E(Y|X = 1) = 3/2$, is larger than the (unconditional) expected value of heads if we do not know what the first flip was, $E(Y) = 1$.

2.3 Useful Properties of Expectations

We will use the following properties throughout the course.

Property 1 (Linearity). If X, Y , and Z are three random variables, and a and b are constants (e.g., numbers like 5 or 8), then:

$$E(aY + bZ) = aE(Y) + bE(Z) \quad (4)$$

and:

$$E(aY + bZ|X) = aE(Y|X) + bE(Z|X) \quad (5)$$

In words, the expectation operator $E(\cdot)$ is linear, that is, we can distribute it over multiplication by constants and summation of random variables.

However, in general, you CANNOT:

- Split up expected values over random variables multiplied together:

$$E(X \cdot Y) \neq E(X) \cdot E(Y) \quad (6)$$

- Distribute or “move inside” the expectation operator if a function is *not* linear. Formally, for any non-linear function $g(X)$:

$$E(g(X)) \neq g(E(X)) \quad (7)$$

Example: $E(X^2) \neq [E(X)]^2$, or $E[e^X] \neq e^{E(X)}$. This only works if the function is linear (as in Property 1).

Property 2 (Law of Iterated Expectations, or LIE). The LIE is:

$$E(Y) = E_X[E(Y|X = x)] \quad (8)$$

In words, we can write the (overall) unconditional mean or expectation of Y , $E(Y)$, as the weighted average over all possible values of X (i.e., as the expectation $E_X(\cdot)$) of the conditional means of Y given a value of X (i.e., $E(Y|X = x)$).

Example: Think about calculating the GPA of a class. In this case, it is possible to write the GPA of a class as:

$$\begin{aligned}\text{The GPA of a Class} &= (\text{GPA for Men})(\text{Fraction of Men}) \\ &\quad + (\text{GPA for Women})(\text{Fraction of Women})\end{aligned}$$

To relate this idea to the LIE formula in (8), let Y = GPA of a Student in the class, and let

$$X = \begin{cases} 1 & \text{if Man} \\ 0 & \text{if Woman} \end{cases}$$

Then, we can write the GPA of a class, $E(Y)$, as:

$$E(Y) = E_X[E(Y|X = x)] = \Pr(X = 1)E(Y|X = 1) + \Pr(X = 0)E(Y|X = 0)$$

Exercise “for fun” (i.e., not to be handed in): Try verifying the LIE for the example we did in Section 2.2. There, we found that $E(Y|X = 1) = 3/2$. Follow similar steps to the ones there to find $E(Y|X = 0)$. Then, find $\Pr(X = 0)$ and $\Pr(X = 1)$. Finally, apply the formula to the right hand of (8) (or the equation in the example about the GPA above). Do you find the same answer we found for $E(Y)$ in Section 2.2? (Spoiler alert: you do.)